



Models for nonlinear Kalman filters

- The linear Kalman filter assumes a system model of the form:

$$\begin{aligned}x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + w_{k-1} \\z_k &= C_k x_k + D_k u_k + v_k.\end{aligned}$$

- But, some systems have significant nonlinearities and cannot be described adequately with this kind of model.
 - Therefore, also, the standard Kalman-filter recursion doesn't apply directly.
- We now generalize to the nonlinear case, with system dynamics described as:

$$\begin{aligned}x_k &= f(x_{k-1}, u_{k-1}, w_{k-1}) \\z_k &= h(x_k, u_k, v_k).\end{aligned}$$

- Functions $f(\cdot)$ and $h(\cdot)$ may be time-varying, but we generally omit the time dependency from the notation for ease of understanding.



Kinds of nonlinear Kalman filters

- There are three basic KF generalizations to nonlinear systems:
 - Extended Kalman filter (EKF): Analytic linearization of model at each point in time: problematic, but still popular.
 - Sigma-point (Unscented) Kalman filter (SPKF/UKF): Statistical/empirical linearization of model at each point in time: can be much better than EKF at same computational complexity.
 - Particle filters: Most precise, but often thousands of times more computations required than either EKF/SPKF.
- This week, we study EKF; next week SPKF.
- Particle filters are the focus of the fourth course.



Prediction steps of nonlinear Kalman filters

- EKF and SPKF both follow same set of general steps as KF.
- Recall general step 1:

Step 1a: State prediction time update

$$\hat{x}_k^- = \mathbb{E}[x_k | \mathbb{Z}_{k-1}] = \mathbb{E}[f(x_{k-1}, u_{k-1}, w_{k-1}) | \mathbb{Z}_{k-1}].$$

Step 1b: Prediction-error covariance time update

$$\Sigma_{\hat{x},k}^- = \mathbb{E}[(\hat{x}_k^-)(\hat{x}_k^-)^T] = \mathbb{E}[(x_k - \hat{x}_k^-)(x_k - \hat{x}_k^-)^T].$$

Step 1c: Predict system output

$$\hat{z}_k = \mathbb{E}[z_k | \mathbb{Z}_{k-1}] = \mathbb{E}[h(x_k, u_k, v_k) | \mathbb{Z}_{k-1}].$$

Prediction

- EKF and SPKF simply have different expressions from KF for evaluating the expectation operations (because of different system model).



Update steps of nonlinear Kalman filters

- EKF and SPKF both follow same set of general steps as KF.
- Recall general step 2:

Step 2a: Estimator gain matrix

$$L_k = \Sigma_{\tilde{x}\tilde{z},k}^{-} \Sigma_{\tilde{z},k}^{-1}$$

Step 2b: State estimate measurement update

$$\hat{x}_k^+ = \hat{x}_k^- + L_k(z_k - \hat{z}_k)$$

Step 2c: Estimation-error covariance measurement update

$$\Sigma_{\tilde{x},k}^+ = \Sigma_{\tilde{x},k}^- - L_k \Sigma_{\tilde{z},k} L_k^T$$

Correction

- Again, EKF and SPKF simply have different expressions from KF for evaluating the expectation operations implied in Step 2a (because of different system model).



Summary

- Systems described by nonlinear models require nonlinear extensions of the Kalman filter to estimate their states.
- We will study two different approaches to generalizing the Kalman-filter equations to nonlinear systems, one leading to EKF and the other to SPKF/UKF.¹
- EKF and SPKF (and variants) still obey predict/correct sequence of six steps.
- So, their derivations will involve evaluating those six steps under different sets of assumptions.

¹cf. http://www.ieeeeghn.org/wiki/index.php/First-Hand:The_Unscented_Transform re. name "unscented."